

Christopher Wang

Curriculum Vitae

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Education

2021–2025 **B.A. Physics, Mathematics**, *Pomona College*, GPA: 3.96

2007–2021 **High School Diploma**, *Hong Kong International School*

Research Interests

Broadly, using mathematical and physical principles to develop a better understanding of learning, memory, prediction, cognition, and intelligence. Statistical and information-theoretic approaches to neural coding and biological sensing; maximum entropy models of complex systems; computational and theoretical neuroscience.

Keywords Artificial Intelligence, Machine Learning, Mechanistic Interpretability, Probability and Information Theory, Applied Mathematics, Dynamical Systems, Reservoir Computing, Computational Mechanics.

Research Experience

Sep '25– **Research Technician**, *Scripps College*

- Applied Maximum Entropy models to neuronal networks, characterizing performance with confusion matrices and a weighted cross-entropy loss to address class imbalance. Selected neuron subsets by predictive contribution to circumvent the curse of dimensionality in partition-function calculations.
- Co-drafted and revised the resulting paper as first author, incorporating collaborator feedback over multiple iterations. Published in *PLOS ONE* on August 11, 2026.
- Independently identified a miscalculation in Professor Sarah Marzen's 2017 *Physical Review E* paper, "Difference between memory and prediction in linear recurrent networks," and drafted its correction. The resulting [erratum was published on July 10, 2026](#).
- Currently investigating, in collaboration with UMD, the link between a reservoir computer's performance (memory and prediction) and how it couples to the underlying data-generation mechanism as viewed in the lens of [computational mechanics](#). Work conducted under the guidance of Professor Sarah Marzen.

2024–2025 **Physics Senior Thesis**, *Pomona College*

Developing and Characterizing a Maximum Entropy Model to Analyze Memory and Prediction in Cultured Cortical Neurons. Advised by Professor Sarah Marzen.

2024–2025 **Mathematics Senior Thesis**, *Pomona College*

A Mathematical Supplement for First-Year Physicists. Advised by Professor Edray Goins.

Summer '23, **Student Researcher**, *Pomona College*

- '24
- Learned general relativity and tensor calculus to model gravitational wave generation of spinning binary systems. Used the Xojo programming language and Git source control to organize and refine a simulator program for the Laser Interferometer Space Antenna (LISA). Resolved theoretical problems in post-Newtonian gravitational wave polarization equations and addressed computing time reduction with extensive quantitative documentation in LaTeX.
 - Cooperated with three other researchers on different parts of the multi-pronged project. Supervised by Professor Thomas Moore.

Summer '22 **Student Researcher**, *Pomona College*

Learned the mathematical basis of chemical kinetics and the kinetic theory of ligand binding (adsorption/desorption). Used GraphPad Prism to model ex vivo drug-screening data on beta-cyclodextrins and derive experimental parameters, contributing to a study on their cholesterol-desorbing competencies (see Publications). Supervised by Professor Malkiat Johal.

Publications

Journal Articles

C. Wang et al., "Analyzing an organism's sensors using Maximum Entropy (MaxEnt) models with bias, variance, and confusion matrices," *PLOS ONE*, vol. 21, no. 8, e0342165, 2026. <https://doi.org/10.1371/journal.pone.0342165>.

J. K. Al-Husseini, E. M. Fong, **C. Wang**, et al., "Ex Vivo Drug Screening Assay with Artificial Membranes: Characterizing Cholesterol Desorbing Competencies of Beta-Cyclodextrins," *Langmuir*, vol. 39, no. 36, pp. 12590–12598, 2023.

Conference Talks and Poster Sessions

C. Wang et al., "Applying Maximum Entropy Models to Organism Sensors," (talk, 20 min, invited paper session) MAA MathFest, Boston, MA, August 7, 2026.

C. Wang et al., "Analyzing an organism's sensors using maximum entropy models with bias, variance, and confusion matrices," (poster) MAA SoCal-Nevada Section Meeting, March 28, 2026.

C. Wang et al., "Analyzing an organism's sensors using maximum entropy models with bias, variance, and confusion matrices," (talk, 10 min) APS March Meeting 2025, Anaheim, CA, 2025.

Honors and Awards

- 2025 Alfred Kwok Memorial Prize, for outstanding leadership and service to the physics department at Pomona College.
- 2024–2025 Sigma Xi (associate member), Claremont Colleges Chapter. Elected on the basis of outstanding aptitude in science as demonstrated by participation in research.
- 2023–2025 Pomona College Scholars, Pomona College. Awarded semesterly to the top quarter of students in each class based on semester GPA.
- 2023 Barry Goldwater Scholarship institutional nominee, Pomona College.
- 2023 Tileston Physics Prize, Pomona College. Awarded to outstanding underclassmen whose record is judged most promising by the physics department.
- 2021 Outstanding Achievement in Mathematics Award, Hong Kong International School (one student per senior class of ~200).
- 2021 National Orchestra Award, Hong Kong International School (one student per senior class of ~200).

Skills

Languages Bilingual fluency in English and Mandarin Chinese.

Coding Python (NumPy, Matplotlib, pandas, SciPy), L^AT_EX, Wolfram Mathematica.

Coursework

- Physics Quantum Mechanics, Classical Mechanics, Statistical Mechanics, Electromagnetism, General Relativity, Particle Physics.
- Math Real and Functional Analysis, Linear and Abstract Algebra, Vector Calculus, Differential Geometry, Point-set and Algebraic Topology, Dynamical Systems, Representation Theory.

Miscellaneous Experience

- Summer '25 **Teaching Assistant, Residential Mentor, *Summer Science Program***
- Served as a teaching assistant and residential mentor for an academic program for rising high-achieving high school seniors centered on orbit determination of a near-Earth asteroid, involving math, computer science, and physics.
 - Set up and maintained residential living areas and facilitated communication between senior faculty and participants. Supported participants' academic and personal needs, via daily faculty debriefs on program and participant progress and longer weekly meetings to reach consensus on each participant's academic and personal well-being.
 - Independently wrote and managed a daily "Question of the Day" to engage students with advanced topics in math, physics, and astronomy.
- Jan '22– **Student Mentor, Teaching Assistant, *Pomona College, Harvey Mudd College***
- May '25
- Assisted students with lab equipment and conducting labs in the fall 2023 and 2024 Foundations of Modern Physics lab course. Gave feedback on lab notebooks, graded lab reports, and guided off-campus field trips. Served as a writing partner at Pomona College's Center for Speaking, Writing, and the Image (CSWIM).
 - Organized weekly mentor sessions for introductory and upper-division Mechanics, Mathematical Methods of Physics, and Vector Calculus.
 - Graded problem sets and gave feedback for Mathematical Methods of Physics, Mechanics, Advanced Linear Algebra, and a first-year graduate quantum mechanics course at Harvey Mudd College.
 - Proofread Professor Stephen R. Garcia's textbook, *Matrix Mathematics: A Second Course in Linear Algebra* (ISBN-13 978-1108837101), providing feedback on mathematical errors and layout.
- Fall '23– **Physics Department Liaison, *Pomona College***
- Spring '25 Facilitated communication between students and faculty in the physics department, including via an anonymous feedback box. Co-organized departmental events, including underclassmen course-registration ice cream socials and departmental retreats and festivals.

Hobbies

Cello, Piano, Photography, Amateur Chemistry, Cooking, Camping, Hiking, Running, Cycling, Badminton.